

Technical Document #383: Blockchain - Comprehensive Overview

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Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Tokenomics studies the economic systems of cryptocurrency tokens.
- Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself.
- Blockchain is a distributed ledger technology that records transactions across many computers.